2. Final exam:
Mathematics for Artificial Intelligence 2

Last name: **First name:**

Matr. nr.:

Linz/Vienna/Bregenz, August 16th, 2022 Signature:

Duration: 180 minutes
Maximum of 58 points

Rules:

- **Put your name and matriculation nr. on every sheet!**
- **Use a new sheet for every exercise!**
- Electronic devices are not allowed!
- Do not use pencil or a red pen.
- The whole procedure of solution is required.
- All results must be simplified as far as possible.

Good luck!

EXERCISE 1: **Limits** (8 points)

Determine if the following proper or improper limits exist and, if applicable, determine the value.

a) $\lim\limits_{x \to 0} \dfrac{e^{2x} - 2x - 1}{x^2}$ b) $\lim\limits_{x \to 1/2} \dfrac{2^{6x}}{3^{4x}}$ c) $\lim\limits_{x \to 0} e^{-1/x}$

Solution: [Points: l'Hospital for a (36p); just putting in for b (2p); argument (e.g. one sided limits) for c (2p); form (1p)]
a) 2
b) 8/9
c) no limit

EXERCISE 2: **Global and Local Extrema** (15 points)

Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x) = \mathrm{e}^{-x^2/2} \cdot x$.

a) Determine all local and global extrema and extreme values of f.

b) Where is f monotonically increasing/decreasing?

c) Where is f convex/concave?

Solution: [Points: f', f'' (3p); solving $f' = 0$ (2p); local max/min (2p); argument for global max/min (2p); values (1p) b) (2p); c) (2p); form (1p)]
We have

$$\begin{aligned} f'(x) &= e^{-x^2/2}(1 - x^2), \\ f''(x) &= e^{-x^2/2}x(x^2 - 3). \end{aligned}$$

a) According to Theorem 5.25 a necessary condition for x being an extreme points is $f'(x) = 0$. This holds for $x_1 = 1$ and $x_2 = -1$. Using Theorem 5.29 we obtain that

$$f''(1) = \frac{-2}{\sqrt{e}} < 0 \qquad \text{and} \qquad f''(-1) = \frac{2}{\sqrt{e}} > 0.$$

Therefore, f has a local minimum at $x_2 = -1$ and a local maximum at $x_1 = 1$. As these are the only local extrema, and $\lim_{x\to\pm\infty} f(x) = 0$, there is a global minimum at $x_2 = -1$ with value $f(x_2) = \frac{-1}{\sqrt{e}}$, and a global maximum at $x_1 = 1$ with value $f(x_1) = \frac{1}{\sqrt{e}}$.

b) The function f is decreasing in $(-\infty, -1) \cup (1, \infty)$ and increasing in $(-1, 1)$.

c) The function f is concave in $[-\infty, -\sqrt{3}] \cup [0, \sqrt{3}]$ and convex in $[-\sqrt{3}, 0] \cup [\sqrt{3}, \infty)$.

Exercise 3: **Taylor series** (7 points)

Determine the Taylor polynomial $T_4(x)$ of order 4 of the function

$$f\colon \mathbb{R} \to \mathbb{R},\ f(x) = x + (\sin(x))^2$$

around $x_0 = \pi$.

Solution: [Points: derivatives & values (3p); Taylor polynomial: "roughly correct" (1p), "$\frac{1}{k!}$" (+1p), "$(? - x_0)$" (+1p); form (1p)]
We have

$$f'(x) = 1 + 2\sin(x)\cos(x), \qquad f''(x) = 2(\cos^2(x) - \sin^2(x)),$$
$$f'''(x) = -8\sin(x)\cos(x), \qquad f^{(4)}(x) = -8(\cos^2(x) - \sin^2(x)),$$

and so

$$f(x_0) = \pi,\ f'(x_0) = 1,\ f''(x_0) = 2,\ f'''(x_0) = 0,\ f^{(4)}(x_0) = -8.$$

Hence,

$$\begin{aligned} T_4(x) &= \pi + (x-\pi) + (x-\pi)^2 - \frac{1}{3}(x-\pi)^4 \\ &= x + (x-\pi)^2 - \frac{1}{3}(x-\pi)^4 \end{aligned}$$

Exercise 4: **Antiderivatives** (8 points)

a) Compute the integral $\int_1^e f_1(x)\,\mathrm{d}x$ for $f_1(x) = 3x^2 \cdot \ln(x^3)$.

b) Find the function f_2 on $(0, \infty)$ with $f_2'(x) = \dfrac{1}{x\ln(x)}$ for all $x \in (0, \infty)$ and $f_2(e) = 1$.

Solution: [Points: a) antiderivative (3p); integral (2p); b) antiderivative (2p); correct value (1p)]

a) We use partial integration with $f'(x) = 3x^2$ and $g(x) = \ln(x^3) = 3\ln(x)$ to find $f(x) = x^3$ and $g'(x) = \frac{3}{x}$, and so

$$\int f_1(x)\mathrm{d}x = 3x^3\ln(x) - \int 3x^2\mathrm{d}x = 3x^3\ln(x) - x^3.$$

Hence,

$$\int_1^e f_1(x)\,\mathrm{d}x \ = \ 2e^3 + 1.$$

b) We use the substitution $t = \ln(x)$, s.t. $dt = \frac{dx}{x}$, and so

$$f_2(x) = \int f_2'(x)\mathrm{d}x = \int \frac{1}{t}\mathrm{d}t = \ln(t) + C = \ln\ln(x) + C.$$

We get $f_2(e) = C$ and so $f_2(x) = 1 + \ln\ln(x)$.

EXERCISE 5: **Fourier series** (10 points)

Compute the Fourier series $Sh(x)$ of the function $h\colon [0,1) \to \mathbb{R}$ with $h(x) = 2x - 1$. Simplify the result! (Complex numbers in canonical form.)
Determine all $x \in [0,1)$ for which $Sh(x)$ equals $h(x)$.

Solution: [Points: Def. $\hat{f}(k)$ (1p); coefficients $k \neq 0$ correct (2p); coefficient $k = 0$ (1p); Sh (2p); convergence $x \neq 0$ (2p); non-convergence for $x = 0$ (1p); simplified (1p)]
We compute the Fourier coefficients according to Definition 7.14:

$$\hat{f}(k) = 2\int_0^1 xe^{-2\pi ikx}dx - \int_0^1 e^{-2\pi ik}dx = -\frac{1}{\pi ik} = \frac{i}{\pi k}$$

for $k \neq 0$. Here we used Lemma 7.15 and Example 7.16. For $k = 0$ we have $\hat{f}(k) = 0$. Thus,

$$Sh(x) = \frac{i}{\pi}\sum_{k\neq 0}\frac{1}{k}e^{2\pi ikx}.$$

Since $h : [0,1) \to \mathbb{R}$ and h is piecewise continuous, we can use Theorem 7.39. (Dirichlet's theorem). Now, h is differentiable in $x \in (0,1)$, and therefore, for those x, $Sh(x) = h(x)$. For $x = 0$ we see that $h(0) = -1$ but

$$Sh(0) = \lim_{n\to\infty} S_n h(0) = \frac{i}{\pi}\sum_{1\leq|k|\leq n}\frac{1}{k} = \frac{i}{\pi}\sum_{k=1}^{n}\left(\frac{1}{k} + \frac{1}{-k}\right) = 0$$

and we do not have convergence.

EXERCISE 6: **Extrema under constraints** (10 points)

Calculate the global extrema of

$$f(x_1, x_2) = 2(x_2 - 1)^2 + (x_1 + 1)^2$$

under the constraint

$$x_1^2 + 2x_2^2 \leq \frac{1}{3}.$$

Solution: [Points: no extrema in Ω^o (2p); finding equations by Lagrange (2p); solving these equations (3p); min/max + values (2p); form (1p)]
First, note that the only stationary point of f on $\mathbb{R}^2$ is at $(x_1, x_2) = (-1, 1)$, which is does not fulfill the constraint. Hence, there are no local/global extrema in $\Omega^o = \{x_1^2 + 2x_2^2 < \frac{1}{3}\}$. We use the method of Lagrange multipliers to find all extrema with $x_1^2 + 2x_2^2 = \frac{1}{3}$. The Lagrange function is

$$\mathcal{L}(x_1, x_2, \lambda) = (x_1 + 1)^2 + 2(x_2 - 1)^2 + \lambda(x_1^2 + 2x_2^2 - 1/3)$$

(with $g(x_1, x_2) = x_1^2 + 2x_2^2$ and $c = 1/3$) and we get

$$\nabla_x \mathcal{L} = (2(x_1 + 1) + \lambda 2x_1, 4(x_2 - 1) + \lambda 4x_2)$$

Now we need to calculate all critical points of $\mathcal{L}$. Since the only critical point of g is $(0,0)$ for which the constraint is not fulfilled, we only have to solve the system

$$\begin{aligned} 2(x_1 + 1) + \lambda 2x_1 &= 0 \\ 4(x_2 - 1) + \lambda 4x_2 &= 0 \\ x_1^2 + 2x_2^2 &= \frac{1}{3}. \end{aligned}$$

The first two equations yield that $\lambda \neq -1$ and

$$x_1 = \frac{-1}{1+\lambda} \qquad \text{and} \qquad x_2 = \frac{1}{1+\lambda},$$

put in the last equation gives

$$\left(\frac{-1}{1+\lambda}\right)^2 + 2\left(\frac{1}{1+\lambda}\right)^2 = \frac{1}{3},$$

yielding $9 = (1+\lambda)^2 \iff 3 = |1+\lambda|$ and resulting in the possible solutions

$$\lambda = 2 \qquad \text{or} \qquad \lambda = -4.$$

With this we have the critical points $(-1/3, 1/3, 2)$ and $(1/3, -1/3, -4)$, and hence that $f(-1/3, 1/3) = 4/3$ is the global minimum and $f(1/3, -1/3) = 16/3 \approx 5.33$ is the global maximum.